

CALCULATION OF PATERNITY PROBABILITIES IN THE HL-A SYSTEM

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Summary: Mayr and Pausch (1975) derived formulae for calculating the proportion of the population from which a presumptive father might come in paternity studies. They ignored the fact that for a certain part of the group of potential fathers the probability is very small that the required genetic material will actually be produced because it can only happen after a translocation. In this paper a different approach is proposed, and the difference in the results of the two sets of formulae illustrated with examples. In view of the importance of these probabilities, e.g. in legal actions, the discrepancies are much too large to be ignored, and it is imperative that the more realistic approach should be adopted.

1. INTRODUCTION

By using two loci in the HL-A system Mayr & Pausch (1975) derived formulae based on the gene - and haplotype frequencies of the population to estimate the proportion of the population from which the natural father might come. The possible haplotype(s) of the natural father is derived from the phenotypes of the mother and child. Mayr & Pausch (1975), in deriving

their formulae, considered transmission of haplotypes from father to child if the alleles were either in cis position (i.e. the alleles transmitted were on the same chromosome) or in the trans position (i.e. the alleles transmitted were on different chromosomes). The reader is referred to Soulier et al. (1974), pp. 250-252 for details. The transmission from a trans position only happens if a cross-over takes place. Mayr (1971, 1972) showed that cross-over inheritance only happens in approximately 0,38% of the times, a value which corresponds well with the values obtained by other authors. This value apparently varies between 0,35% and 0,80%.

In the paternity probability defined in this paper, provision is made (by law of total probability) for the possibilities of inheritance of alleles from either the cis or the trans position. Mayr & Pausch, on the other hand, calculated FvM, the fraction of the population from which the natural father might come, and defined the probability of exclusion as $VACH = 1 - FvM$. They used the fractions due to cis and trans inheritance without weighting them according to their probability of occurring, thus ignoring the important fact that the probability of a cross-over taking place is at most 0,8%. In the examples at the end of this paper it is seen that in certain instances there is a marked difference between the probabilities of exclusion obtained by Mayr and Pausch and the paternity probabilities derived in this paper. We claim that paternity probabilities are more useful than probabilities of exclusion.

2. DERIVATION OF PROBABILITIES

Consider two loci A and B on a chromosome. At each locus any one of several alleles of the A gene, indicated by $A_1, A_2, \dots, A_m$, or $B_1, B_2, \dots, B_n$ of the B gene, may occur. The corresponding probabilities of the alleles occurring at A and B loci are $p_1, p_2, \dots, p_n$ respectively. As it is

not always possible to identify the alleles present, provision is made for unknown alleles A_x and B_x with probabilities p_x and $p_{.x}$ respectively. The haplotypes are indicated by $A_1B_1, A_1B_2, \dots, A_xB_x$ with corresponding probabilities $p_{11}, p_{12}, \dots, p_{xx}$ of occurring on a chromosome. It should be noted that

$$\sum_{i=1}^x p_{ij} = p_{.j}, \quad \sum_{j=1}^x p_{ij} = p_{i.} \quad \text{and} \quad \sum_{i=1}^x p_{i.} = \sum_{j=1}^x p_{.j} = 1$$

where $\sum_{i=1}^x$ implies that the index i sums over the values $1, 2, \dots, m$ and x . Similarly $j = 1, 2, \dots, n, x$.

Let $P(\text{cis}) = c$ be the probability of a haplotype in the cis position being transmitted and $P(\text{trans}) = t = 1 - c$ the probability of a haplotype in the trans position being transmitted (obviously $P(\text{cis}) + P(\text{trans}) = 1$ because these are the only possibilities).

Let $P_c(A_kB_q)$ be the proportion of men who could contribute A_kB_q from the cis position, and $P_t(A_kB_q)$ the proportion who could contribute A_kB_q from the trans position. We define $F(A_kB_q) = cP_c(A_kB_q) + tP_t(A_kB_q)$, the *weighted* proportion of potential contributors of A_kB_q . We claim that $1 - F(A_kB_q)$, the *weighted probability of exclusion*, would be a more realistic criterion than $VACH = 1 - FvM$, the probability of exclusion of Mayr & Pausch. We will, however, go a step further towards a probability that takes inheritance probabilities into account also. Before doing this, we first calculate basic probabilities.

The diagram in table 1 illustrates which probabilities are contained in the cis and trans cases. Consider two loci A and B , with three alleles $1, 2$ and x at A and four alleles $1, 2,$

TABLE 1. Combinations of chromosomes in parents which may result in the haplotype A_1B_1 being transmitted in cis(c) or trans(t) position, with corresponding probabilities.

CHROMOSOME 2

HAPLOTYPE PAIR (PROB- ABILITY)	$A_1B_1(p_{11})$	$A_1B_2(p_{12})$	$A_1B_3(p_{13})$	$A_1B_x(p_{1x})$	$A_2B_1(p_{21})$	$A_2B_2(p_{22})$	$A_2B_3(p_{23})$	$A_2B_x(p_{2x})$	$A_xB_1(p_{x1})$	$A_xB_2(p_{x2})$	$A_xB_3(p_{x3})$	$A_xB_x(p_{xx})$	TOTAL
$A_1B_1(p_{11})$	ct p_{1111}	ct p_{1112}	ct p_{1113}	ct p_{111x}	ct p_{1121}	c p_{1122}	c p_{1123}	c p_{112x}	ct p_{11x1}	c p_{11x2}	c p_{11x3}	c p_{11xx}	$p_{11..}$
$A_1B_2(p_{12})$	ct p_{1211}	p_{1212}	p_{1213}	p_{121x}	t p_{1221}	p_{1222}	p_{1223}	p_{122x}	t p_{12x1}	p_{12x2}	p_{12x3}	p_{12xx}	$p_{12..}$
$A_1B_3(p_{13})$	ct p_{1311}	p_{1312}	p_{1313}	p_{131x}	t p_{1321}	p_{1322}	p_{1323}	p_{132x}	t p_{13x1}	p_{13x2}	p_{13x3}	p_{13xx}	$p_{13..}$
$A_1B_x(p_{1x})$	ct p_{1x11}	p_{1x12}	p_{1x13}	p_{1x1x}	t p_{1x21}	p_{1x22}	p_{1x23}	p_{1x2x}	t p_{1xx1}	p_{1xx2}	p_{1xx3}	p_{1xxx}	$p_{1x..}$
$A_2B_1(p_{21})$	ct p_{2111}	t p_{2112}	t p_{2113}	t p_{211x}	p_{2121}	p_{2122}	p_{2123}	p_{212x}	p_{21x1}	p_{21x2}	p_{21x3}	p_{21xx}	$p_{21..}$
$A_2B_2(p_{22})$	c p_{2211}	p_{2212}	p_{2213}	p_{221x}	p_{2221}	p_{2222}	p_{2223}	p_{222x}	p_{22x1}	p_{22x2}	p_{22x3}	p_{22xx}	$p_{22..}$
$A_2B_3(p_{23})$	c p_{2311}	p_{2312}	p_{2313}	p_{231x}	p_{2321}	p_{2322}	p_{2323}	p_{232x}	p_{23x1}	p_{23x2}	p_{23x3}	p_{23xx}	$p_{23..}$
$A_2B_x(p_{2x})$	c p_{2x11}	p_{2x12}	p_{2x13}	p_{2x1x}	p_{2x21}	p_{2x22}	p_{2x23}	p_{2x2x}	p_{2xx1}	p_{2xx2}	p_{2xx3}	p_{2xxx}	$p_{2x..}$
$A_xB_1(p_{x1})$	ct p_{x111}	t p_{x112}	t p_{x113}	t p_{x11x}	p_{x121}	p_{x122}	p_{x123}	p_{x12x}	p_{x1x1}	p_{x1x2}	p_{x1x3}	p_{x1xx}	$p_{x1..}$
$A_xB_2(p_{x2})$	c p_{x211}	p_{x212}	p_{x213}	p_{x21x}	p_{x221}	p_{x222}	p_{x223}	p_{x22x}	p_{x2x1}	p_{x2x2}	p_{x2x3}	p_{x2xx}	$p_{x2..}$
$A_xB_3(p_{x3})$	c p_{x311}	p_{x312}	p_{x313}	p_{x31x}	p_{x321}	p_{x322}	p_{x323}	p_{x32x}	p_{x3x1}	p_{x3x2}	p_{x3x3}	p_{x3xx}	$p_{x3..}$
$A_xB_x(p_{xx})$	c p_{xx11}	p_{xx12}	p_{xx13}	p_{xx1x}	p_{xx21}	p_{xx22}	p_{xx23}	p_{xx2x}	p_{xxx1}	p_{xxx2}	p_{xxx3}	p_{xxxx}	$p_{xx..}$
TOTAL	$p_{..11}$	$p_{..12}$	$p_{..13}$	$p_{..1x}$	$p_{..21}$	$p_{..22}$	$p_{..23}$	$p_{..2x}$	$p_{..x1}$	$p_{..x2}$	$p_{..x3}$	$p_{..xx}$	$p_{....} = 1$

CHROMOSOME 1

3 and x at B. The two-way illustration gives all possible combinations of the haplotypes on the two chromosomes in the father. Assume too, that the child has inherited the combination A_1B_1 from the natural, but unknown father. The possible combinations of haplotype pairs which could result in a transmission of A_1B_1 to the child are indicated in table 1 by c for cis or t for trans position.

The indices k and ℓ are used for the A locus only and they vary from q to m and x . The indices q and r are used for the B locus and can vary from 1 to n and x . To calculate the probabilities (we prefer to talk about probabilities rather than proportions), we introduce the following notation:

$P(A_kB_q \text{ on chromosome 1 and } A_\ell B_r \text{ on chromosome 2}) = p_{kq\ell r}$.

For illustration purposes the values of $p_{kq\ell r}$ are given in

table 1. We easily get $P_c(A_kB_q) = p_{kq..} + p_{..kq} - p_{kqkq}$

which contains the probabilities in the row A_kB_q and the

column A_kB_q , with the probability common to both, being sub-

tracted to avoid duplication. The advantage of our notation is that we can get the expression for the inheritance from the trans position simply by interchanging the indices in the second and fourth positions, e.g. $P_t(A_kB_q) = p_{k..q} + p_{.qk.} - p_{kqkq}$.

Similarly, using well-known set notation, we have:

$$\begin{aligned} P_c\{A_k \cap (B_q \cup B_r)\} &= P_c\{A_k \cap B_q\} + P_c\{A_k \cap B_r\} - P_c\{(A_k \cap B_q) \cap (A_k \cap B_r)\} \\ &= p_{kq..} + p_{..kq} - p_{kqkq} + p_{kr..} + p_{..kr} - p_{krkr} \\ &\quad - p_{kqkr} - p_{krkq} \end{aligned}$$

and by changing indices

$$\begin{aligned} P_t\{A_k \cap (B_q \cup B_r)\} &= p_{k..q} + p_{.qk.} - p_{kqkq} + p_{k..r} + p_{.rk.} - p_{krkr} \\ &\quad - p_{krkq} - p_{kqkr} \end{aligned}$$

As we do not know the genotypes, but only the phenotypes, it is necessary to transform to our initial notation. We get:

$$P_c(A_k B_q) = p_{kq} + p_{kq} - p_{kq}^2 = 2p_{kq} - p_{kq}^2 = 2u - u^2 \quad \text{and}$$

$$P_t(A_k B_q) = p_k \cdot p_q + p_q \cdot p_k - p_{kq}^2 = 2p_k \cdot p_q - p_{kq}^2 = 2p_k \cdot p_q - u^2$$

with $u = p_{kq}$ (u is used as a generic constant).

All expressions have been obtained by means of the extended notation (cf. Reinach & Lemmer, 1980) and the final expressions for all possible cases are given in the appendix in the initial notation. It must be pointed out that the extended notation was used to calculate the probabilities in a logical way, but does not imply that the genotypes are assumed to be known.

We now go a step further and define as *parentage probability* of the group of potential fathers:

$$\begin{aligned} P(A_k B_q) &= P(A_k B_q \text{ is contributed by a randomly drawn man from the population}) \\ &= cP(A_k B_q \text{ contributed from the cis position}) \\ &\quad + tP(A_k B_q \text{ contributed from the trans position}) \text{ by the law of total probability.} \end{aligned}$$

$$\begin{aligned} \text{Now } P'_c(A_k B_q) &\equiv P(A_k B_q \text{ contributed from the cis position}) \\ &= \frac{1}{2}P_c(A_k B_q) + \frac{1}{2}p_{kq}^2 \end{aligned}$$

because

- (i) $P_c(A_k B_q)$ = probability that a randomly drawn man belongs to the group of potential fathers who have $A_k B_q$ in cis position on at least one chromosome.
- (ii) The probability that a man, who has $A_k B_q$ on exactly one chromosome, will contribute $A_k B_q$, is $\frac{1}{2}$.

- (iii) The group of potential fathers referred to in (i) includes those who have $A_k B_q$ on *both* chromosomes, and for such a father the probability is 1 (not $\frac{1}{2}$) that $A_k B_q$ will be contributed.

Thus a "correction factor" $\frac{1}{2}p_{kq}^2$ must be added to $\frac{1}{2}P_c(A_k B_q)$.

Similarly for the trans case,

$$\begin{aligned} P'_t(A_k B_q) &\equiv P(A_k B_q \text{ transmitted from the trans position}) \\ &= \frac{1}{2}P_t(A_k B_q) + \frac{1}{2}p_{kq}^2. \end{aligned}$$

The required parentage probability of the *group of potential fathers* is

$$\begin{aligned} P(A_k B_q) &= cP'_c(A_k B_q) + tP'_t(A_k B_q) \\ &= c[\frac{1}{2}P_c(A_k B_q) + \frac{1}{2}Qu] + t[\frac{1}{2}P_t(A_k B_q) + \frac{1}{2}Qu] \text{ with } Qu \text{ the} \\ &\quad \text{correction factor, which is in general equal to} \\ &\quad \text{the sum of the purely quadratic element(s)* of } u^2, \\ &\quad \text{where } u \text{ is a generic constant used in the appen-} \\ &\quad \text{dix and equal to } p_{kq} \text{ in the present case} \\ &= \frac{1}{2}cP_c(A_k B_q) + \frac{1}{2}tP_t(A_k B_q) + \frac{1}{2}Qu \\ &= c[\frac{1}{2}(2p_{kq} - p_{kq}^2) + \frac{1}{2}p_{kq}^2] + t[\frac{1}{2}(2p_{k.p.q} - p_{kq}^2) + \frac{1}{2}p_{kq}^2] \\ &= cp_{kq} + tp_{k.p.q}. \end{aligned}$$

Note that for any haplotype the probability is

$$P(\cdot) = \frac{1}{2}cP_c(\cdot) + \frac{1}{2}tP_t(\cdot) + \frac{1}{2}Qu.$$

If the probability $P(A_k B_q)$ is small, say 0.05, and the alleged father falls in this small group of fathers who *could produce this combination*, then it is taken as a strong indication that he could have been the father. The probability $1 - P(A_k B_q)$ is what we call the *paternity probability* (of the alleged father) and we consider it as more suitable than VACH of Mayr & Pausch.

* If $u = p_{kq} + p_{kr}$, $Qu = p_{kq}^2 + p_{kr}^2$. If $u = p_{kq} + p_{kx} + p_{xq} + p_{xx}$,

$$Qu = p_{kq}^2 + p_{kx}^2 + p_{xq}^2 + p_{xx}^2.$$

3. ILLUSTRATIONS

In this paragraph the paternity probabilities are compared with the probabilities of exclusion of Mayr & Pausch. The probability of a cross-over is taken as 0,008, and it is reasonable to assume that the Hardy-Weinberg assumptions hold for the population from which the gene probabilities (frequencies) are estimated. These assumptions are that the population from which the gene probabilities are estimated is large, that random mating takes place and that no gene drift takes place, i.e. the genes are in equilibrium.

In order to show how Mayr & Pausch calculated their probabilities, we consider *their* calculation of FvM for A_1B_1 :

$$\begin{aligned}
 FvM(A_1B_1) &= (p_{1122} + p_{1123} + p_{112x} + p_{11x3} + p_{11xx} + p_{2211} + p_{2311} \\
 &\quad + p_{2x11} + p_{x211} + p_{x311} + p_{xx11}) + (p_{1111} + p_{1112} + p_{1113} \\
 &\quad + p_{111x} + p_{1121} + p_{11x1} + p_{1211} + p_{1311} + p_{1x11} + p_{2111} \\
 &\quad + p_{x111}) + (p_{1221} + p_{12x1} + p_{1321} + p_{13x1} + p_{1x21} + p_{1xx1} \\
 &\quad + p_{2112} + p_{2113} + p_{211x} + p_{x112} + p_{x113} + p_{x11x}) \\
 &= P_1 + P_2 + P_3 \quad \text{with } P_1 \text{ the expression in the} \\
 &\quad \text{first set of brackets, etc.}
 \end{aligned}$$

Now $P_1 + P_2$ consists of all the terms that appear in the cis transmission, i.e. $P_1 + P_2 = P_c(A_1B_1)$. Mayr & Pausch simply added the terms in P_3 to $P_1 + P_2$. Actually, $P_t(A_1B_1) = P_2 + P_3$ as can clearly be seen from table 1, and the correct form of $F(A_1B_1)$ is, as we have mentioned before:

$$\begin{aligned}
 F(A_1B_1) &= cP_c(A_1B_1) + (1-c)P_t(A_1B_1) \\
 &= c(P_1 + P_2) + (1-c)(P_2 + P_3) \\
 &= cP_1 + P_2 + (1-c)P_3
 \end{aligned}$$

which clearly differs from $P_1 + P_2 + P_3$ of Mayr & Pausch. Whatever the value of c , $F(A_1B_1) \leq P_1 + P_2 + P_3 \equiv FvM(A_1B_1)$. Thus the weighted probability of exclusion is always $\geq$ the probability of exclusion. Unfortunately, it is not easy to compare the paternity probability (of the alleged father) and the probability of exclusion of Mayr & Pausch mathematically, so we do it for some true cases.

Tables for haplotype and gene probabilities were developed by Reinach (1980) and are based on Caucasian persons in Johannesburg, South Africa. Using these tables the exclusion probabilities are calculated for several haplotype combinations and are given in table 2.

TABLE 2 - Calculation of exclusion probabilities of Mayr & Pausch as well as Paternity Probabilities assuming that the probability of a cross-over taking place is 0,008 in the HLA-system.

Possible allele Combinations of father*	Proportion of population where genes are in		Exclusion Probability M & P	Paternity Probability R & L
	Cis	Trans		
$A_9 \cap B_8$	0,0002	0,0239	0,9759	0,9998
$A_9 \cap (B_7 \cup B_8)$	0,0246	0,0483	0,9270	0,9875
$(A_2 \cup A_9) \cap (B_7 \cup B_8)$	0,1187	0,0996	0,7817	0,9396
$(A_2 \cup A_9) \cap (B_8 \cup B_x)$	0,0748	0,0693	0,8560	0,9623
$(A_9 \cup A_x) \cap (B_8 \cup B_x)$	0,0632	0,0272	0,9096	0,9683
$(A_1 \cup A_x) \cap B_{35}$	0,0394	0,0236	0,9370	0,9802
$A_2 \cap B_{27}$	0,0310	0,0185	0,9505	0,9844

*The numbers correspond to the number of the alleles at the A and B loci of the HL-A system.

From table 2 it is seen that the authors' formulae result in paternity probabilities which are always higher than the exclusion probabilities obtained by Mayr & Pausch.

4. CONCLUSIONS

Paternity probabilities, as defined in this paper, are undoubtedly more useful than probabilities of exclusion of Mayr & Pausch. Moreover, if the formulae of Mayr and Pausch are used to calculate an exclusion probability, the danger is that it may be interpreted as a paternity probability, thus obtaining an under-estimation of the true probability.

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APPENDIX

$$(a) P_t(A_k \cap B_q) = 2p_k \cdot p_q - u^2$$

$$P_c(A_k \cap B_q) = 2u - u^2 \quad \text{with } u = p_{kq}$$

$$(b) P_t\{A_k \cap (B_q \cup B_r)\} = 2p_k \cdot (p_q + p_r) - u^2$$

$$P_c\{A_k \cap (B_q \cup B_r)\} = 2u - u^2 \quad \text{with } u = p_{kq} + p_{kr}$$

$$(c) P_t\{A_k \cap (B_q \cup B_x)\} = 2p_k \cdot (p_q + p_x) - u^2 + \sum_{j \neq q, x} (2p_j p_{kj} - p_{kj}^2)$$

$$P_c\{A_k \cap (B_q \cup B_x)\} = 2u - u^2 + \sum_{j \neq q, x} (2p_j p_{kj} - p_{kj}^2) \quad \text{with } u = p_{kq} + p_{kx}$$

$$(d) P_t(A_k \cap B_x) = 2p_k \cdot p_x - u^2 + \sum_{j \neq x} (2p_j p_{kj} - p_{kj}^2)$$

$$P_c(A_k \cap B_x) = 2u - u^2 + \sum_{j \neq x} (2p_j p_{kj} - p_{kj}^2) \quad \text{with } u = p_{kx}$$

$$(e) P_t\{(A_k \cup A_\ell) \cap (B_q \cup B_r)\} = 2(p_k + p_\ell) \cdot (p_q + p_r) - u^2$$

$$P_c\{(A_k \cup A_\ell) \cap (B_q \cup B_r)\} = 2u - u^2 \quad \text{with } u = p_{kq} + p_{kr} + p_{\ell q} + p_{\ell r}$$

$$(f) P_t\{(A_k \cup A_\ell) \cap (B_q \cup B_x)\} = 2(p_k + p_\ell) \cdot (p_q + p_x) - u^2$$

$$+ 2 \sum_{j \neq q, x} p_j (p_{kj} + p_{\ell j})$$

$$- \sum_{j \neq q, x} (p_{kj} + p_{\ell j})^2$$

$$P_c\{(A_k \cup A_\ell) \cap (B_q \cup B_x)\} = 2u - u^2 + 2 \sum_{j \neq q, x} p_j (p_{kj} + p_{\ell j})$$

$$- \sum_{j \neq q, x} (p_{kj} + p_{\ell j})^2$$

$$\text{with } u = p_{kq} + p_{kx} + p_{\ell q} + p_{\ell x}$$

$$(g) P_t\{(A_k \cup A_\ell) \cap B_x\} = 2p_x (p_k + p_\ell) - u^2 + 2 \sum_{j \neq x} p_j (p_{kj} + p_{\ell j})$$

$$- \sum_{j \neq x} (p_{kj} + p_{\ell j})^2$$

$$P_c\{(A_k \cup A_\ell) \cap B_x\} = 2u - u^2 + 2\sum_{j \neq x} p_{.j}(p_{kj} + p_{\ell j}) - \sum_{j \neq x} (p_{kj} + p_{\ell j})^2$$

$$\text{with } u = p_{kx} + p_{\ell x}$$

$$\begin{aligned} (h) \quad P_t\{(A_k \cup A_x) \cap (B_q \cup B_x)\} &= 2(p_{k.} + p_{x.})(p_{.q} + p_{.x}) \\ &\quad - u^2 + 2\sum_{i \neq k, x} p_{i.}(p_{iq} + p_{ix}) \\ &\quad - \sum_{i \neq k, x} (p_{iq} + p_{ix})^2 + 2\sum_{j \neq q, x} p_{.j}(p_{kj} + p_{xj}) \\ &\quad - \sum_{j \neq q, x} (p_{kj} + p_{xj})^2 + \sum_{i \neq k, x} \sum_{j \neq q, x} p_{ij}^2 \end{aligned}$$

$$\begin{aligned} P_c\{(A_k \cup A_x) \cap (B_q \cup B_x)\} &= 2u - u^2 + 2\sum_{i \neq k, x} p_{i.}(p_{iq} + p_{ix}) \\ &\quad - \sum_{i \neq k, x} (p_{iq} + p_{ix})^2 + 2\sum_{j \neq q, x} p_{.j}(p_{kj} + p_{xj}) \\ &\quad - \sum_{j \neq q, x} (p_{kj} + p_{xj})^2 + \sum_{i \neq k, x} \sum_{j \neq q, x} p_{ij}^2 \end{aligned}$$

$$\text{with } u = p_{kq} + p_{kx} + p_{xq} + p_{xx}$$

$$\begin{aligned} (i) \quad P_t\{(A_k \cup A_x) \cap B_x\} &= 2p_{.x}(p_{k.} + p_{x.}) - u^2 + \sum_{i \neq k, x} p_{i.}p_{ix} - \sum_{i \neq k, x} p_{ix}^2 \\ &\quad + 2\sum_{j \neq x} p_{.j}(p_{kj} + p_{xj}) - \sum_{j \neq x} (p_{kj} + p_{xj})^2 + \sum_{i \neq k, x} \sum_{j \neq x} p_{ij}^2 \end{aligned}$$

$$\begin{aligned} P_c\{(A_k \cup A_x) \cap B_x\} &= 2u - u^2 + 2\sum_{i \neq k, x} p_{i.}p_{ix} - \sum_{i \neq k, x} p_{ix}^2 + 2\sum_{j \neq x} p_{.j}(p_{kj} + p_{xj}) \\ &\quad - \sum_{j \neq x} (p_{kj} + p_{xj})^2 + \sum_{i \neq k, x} \sum_{j \neq x} p_{ij}^2 \end{aligned}$$

$$\text{with } u = p_{kx} + p_{xx}$$

$$\begin{aligned} (j) \quad P_t(A_x \cap B_x) &= 2p_{x.}p_{.x} - u^2 + \sum_{i \neq x} (2p_{i.}p_{ix} - p_{ix}^2) + \sum_{j \neq x} (2p_{.j}p_{xj} - p_{xj}^2) \\ &\quad + \sum_{i \neq x} \sum_{j \neq x} p_{ij}^2 \end{aligned}$$

$$P_c(A_x \cap B_x) = 2u - u^2 + \sum_{i \neq x} (2p_i \cdot p_{ix} - p_{ix}^2) + \sum_{j \neq x} (2p_j \cdot p_{xj} - p_{xj}^2) \\ + \sum_{i \neq x} \sum_{j \neq x} p_{ij}^2 \quad \text{with } u = p_{xx}$$

$$(k) P_t\{(A_k \cup A_\ell) \cap B_q\} = 2p_q(p_k + p_\ell) - u^2$$

$$P_c\{(A_k \cup A_\ell) \cap B_q\} = 2u - u^2 \quad \text{with } u = p_{kq} + p_{\ell q}$$

$$(l) P_t\{A_k \cup A_x \cap B_q\} = 2p_q(p_k + p_x) - u^2 + \sum_{i \neq k, x} p_{iq}(2p_i - p_{iq})$$

$$P_c\{A_k \cup A_x \cap B_q\} = 2u - u^2 + \sum_{i \neq k, x} p_{iq}(2p_i - p_{iq}) \quad \text{with } u = p_{kq} + p_{xq}$$

$$(m) P_t\{A_k \cup A_x \cap (B_q \cup B_r)\} = 2(p_k + p_x)(p_q + p_r) - u^2 \\ + 2 \sum_{i \neq k, x} p_i(p_{iq} + p_{ir}) - \sum_{i \neq k, x} (p_{iq} + p_{ir})^2$$

$$P_c\{(A_k \cup A_x) \cap (B_q \cup B_r)\} = 2u - u^2 + 2 \sum_{i \neq k, x} p_i(p_{iq} + p_{ir}) - \sum_{i \neq k, x} (p_{iq} + p_{ir})^2$$

$$\text{with } u = p_{kq} + p_{kr} + p_{xq} + p_{xr}$$

$$(n) P_t(A_x \cap B_q) = 2p_x \cdot p_q - u^2 + \sum_{i \neq x} p_{iq}(2p_i - p_{iq})$$

$$P_c(A_x \cap B_q) = 2u - u^2 + \sum_{i \neq x} p_{iq}(2p_i - p_{iq}) \quad \text{with } u = p_{xq}$$

$$(o) P_t\{A_x \cap (B_q \cup B_r)\} = 2p_x(p_q + p_r) - u^2 + 2 \sum_{i \neq x} p_i(p_{iq} + p_{ir}) \\ - \sum_{i \neq x} (p_{iq} + p_{ir})^2$$

$$P_c\{A_x \cap (B_q \cup B_r)\} = 2u - u^2 + 2 \sum_{i \neq x} p_i(p_{iq} + p_{ir}) - \sum_{i \neq x} (p_{iq} + p_{ir})^2$$

$$\text{with } u = p_{xq} + p_{xr}$$

$$\begin{aligned}
(p) \quad P_t\{A_x \cap (B_q \cup B_x)\} &= 2p_x \cdot (p_{\cdot q} + p_{\cdot x}) - u^2 + 2 \sum_{i \neq x} p_i \cdot (p_{iq} + p_{ix}) \\
&\quad - \sum_{i \neq x} (p_{iq} + p_{ix})^2 + 2 \sum_{j \neq q, x} p_{\cdot j} p_{xj} - \sum_{j \neq q, x} p_{xj}^2 \\
&\quad + \sum_{j \neq q, x} \sum_{i \neq x} p_{ij}^2 \\
P_c\{A_x \cap (B_q \cup B_x)\} &= 2u - u^2 + 2 \sum_{i \neq x} p_i \cdot (p_{iq} + p_{ix}) - \sum_{i \neq x} (p_{iq} + p_{ix})^2 \\
&\quad + 2 \sum_{j \neq q, x} p_{\cdot j} p_{xj} + \sum_{j \neq q, x} p_{xj}^2 + \sum_{j \neq q, x} \sum_{i \neq x} p_{ij}^2
\end{aligned}$$

with $u = p_{xq} + p_{xx}$.

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